SIMATS ENGINEERING

ASSESSMENT TOOL 3

Annexure A – Descriptive Written Test

Course Name	Artificial Intelligence	Course Code	CSA17
Assessment	Tool 3 – Descriptive Written Test	Weightage	30%
Course Outcome	CO1 (BL3)	Duration / Marks	45 Minutes / 10 Marks

CO1: *Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)*

Code of Conduct

I **M. Vishnu Vardhan Reddy /192325061 (Name / Reg No)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

M.V.V.R

Signature of the student:

Question 1 (10 Marks)

> A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Answer begins on the next page.

1. Introduction: The Smart Home as an Intelligent Agent

An **intelligent agent** is any entity that **perceives** its environment through **sensors** and **acts** upon that environment through **actuators** in order to achieve its objectives. Mathematically, the behaviour of an agent is described by the **agent function** f : P* → A, which maps the complete percept history to an action. A **rational agent** chooses, for every percept sequence, the action expected to maximise its **performance measure**, given the knowledge it has.

A smart home controller fits this definition exactly. It continuously runs a **perceive → decide → act** cycle: motion, temperature, light and door sensors supply percepts; lights, air-conditioner, locks and alarms are the actuators; and the controller software is the agent program (Figure 1).

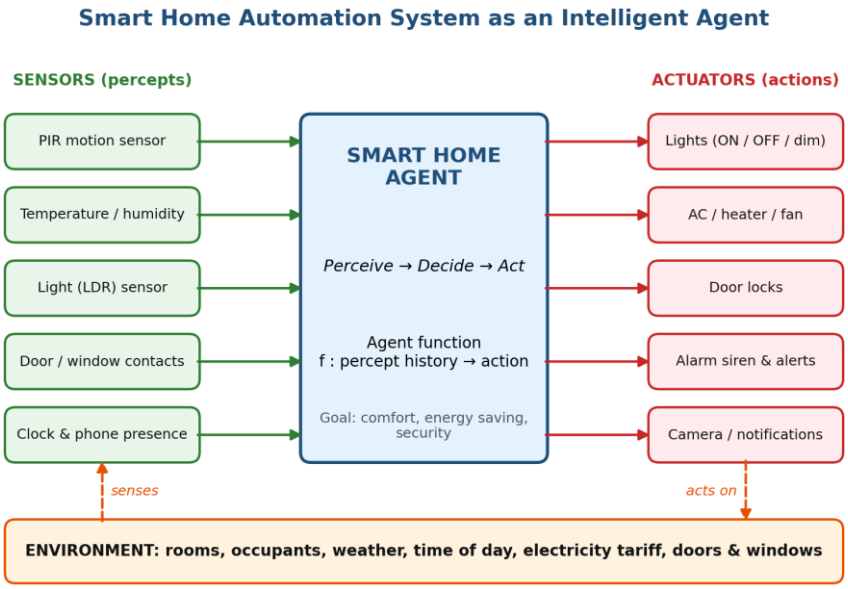


Figure 1: Smart home automation system modelled as an intelligent agent

1.1 PEAS Description of the Smart Home Agent

Before choosing an agent type, the task environment is specified using the PEAS framework:

PEAS element	Smart home specification
Performance measure	Occupant comfort (temperature and lighting as preferred), low energy consumption and cost, security (intrusions detected, few false alarms), fast response time, safety (smoke / fire).
Environment	Rooms of the house, occupants and visitors, weather, time of day, electricity tariff, doors and windows.
Actuators	Smart lights (ON / OFF / dim), AC / heater / fan, door locks, alarm siren, camera control, mobile notifications.
Sensors	PIR motion sensors, temperature and humidity sensors, light (LDR) sensor, door / window contact sensors, smoke detector, camera, clock, smartphone presence.

1.2 Properties of the Environment

The nature of the environment decides how sophisticated the agent must be:

Property	Smart home	Reason
Observability	Partially observable	A PIR sensor cannot see a person sitting still; the agent cannot know who is in which room with certainty.
Determinism	Stochastic	User behaviour, weather and sensor noise are unpredictable.
Episodic / Sequential	Sequential	Current decisions affect the future (e.g. switching on the AC now changes the room temperature later).
Static / Dynamic	Dynamic	The environment changes while the agent is still deliberating.
Discrete / Continuous	Mixed	Temperature and light are continuous; device states (ON / OFF) are discrete.
Agents	Multi-agent	Several family members have different, sometimes conflicting preferences.

Because the environment is partially observable, stochastic, dynamic and multi-agent, **no single simple design is enough**, which motivates the comparison of agent types below.

2. Types of Intelligent Agents and Their Role in the Smart Home

2.1 Simple Reflex Agent

Concept. A simple reflex agent selects actions **only on the basis of the current percept**, ignoring percept history. It uses **condition–action (IF–THEN) rules**. It works correctly only when the right decision can be made from the present percept alone, that is, when the environment is fully observable.

```
function SIMPLE-REFLEX-AGENT(percept):
    state ← INTERPRET-INPUT(percept)
    rule  ← RULE-MATCH(state, rules)
    return rule.ACTION
```

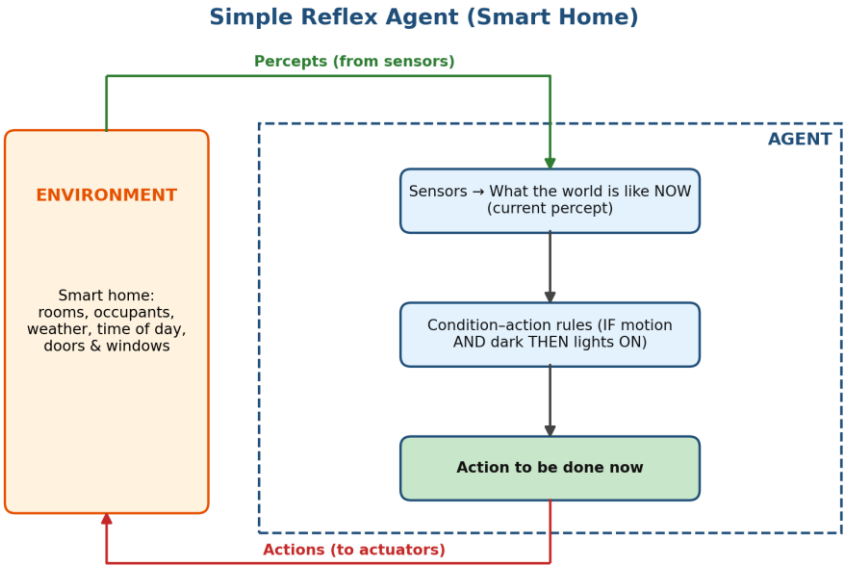


Figure 2: Simple reflex agent for the smart home

Working in the smart home. The agent reacts instantly to what its sensors report right now:

- IF motion is detected AND the room is dark THEN turn the lights ON.
- IF smoke is detected THEN sound the alarm and unlock exit doors.
- IF a door opens AND the alarm is armed THEN trigger the siren.
- IF temperature > 30 °C THEN switch the AC ON.

Advantages: very fast, simple, cheap, predictable and easy to verify, which makes it ideal for safety-critical instant responses.

Limitations: it has no memory, so it cannot tell that a person is still in a room once motion stops (lights may switch off while someone watches TV). It cannot use time-of-day patterns, cannot learn, may oscillate around a threshold (AC switching ON/OFF repeatedly), cannot resolve rule conflicts, and fails under partial observability.

2.2 Model-based Reflex Agent

Concept. A model-based reflex agent maintains an **internal state** that summarises the percept history, and uses a **model of the world** with two parts: (i) **how the world evolves** independently of the agent, and (ii) **what the agent's actions do**. This lets it handle partially observable environments.

```
function MODEL-BASED-REFLEX-AGENT(percept):
    state ← UPDATE-STATE(state, last_action, percept, model)
    rule  ← RULE-MATCH(state, rules)
    return rule.ACTION
```

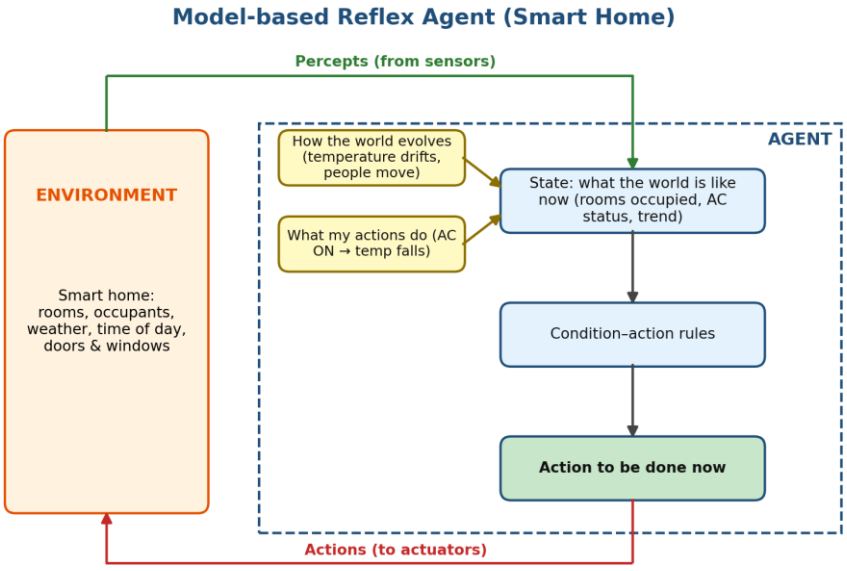


Figure 3: Model-based reflex agent for the smart home

Working in the smart home. The agent keeps a running picture of the house:

- **Occupancy tracking:** a person entered the living room 3 minutes ago and no one left, so the room is still occupied even though the PIR sensor is silent.
- **Device status:** it remembers that the AC is ON and a window is open, and so switches the AC off to avoid wasting energy.
- **Trend model:** temperature is rising 0.5 °C every 10 minutes, so it predicts the room will become uncomfortable and acts early.
- **Presence and security state:** it knows whether the alarm is armed and whether residents' phones are at home.

Rules now use state instead of raw percepts, for example: IF room_unoccupied for 15 min THEN lights OFF; IF window_open AND AC_on THEN AC OFF.

Advantages: copes with partial observability and noisy sensors, avoids flickering, and gives context-aware behaviour.

Limitations: still purely reactive. It has no explicit objective, cannot compare alternative futures, its rules are hand-written, and an inaccurate model gives wrong decisions.

2.3 Goal-based Agent

Concept. A goal-based agent has explicit **goal information** that describes desirable situations. It asks "what will happen if I do action A?" and uses **search and planning** to find a sequence of actions that reaches the goal. Behaviour can be changed simply by changing the goal, without rewriting rules.

Planning is formulated as a **state-space search** problem, which links directly to the uninformed search methods of CO1: **states** are device configurations and room conditions, **actions** are device operations, the **goal test** checks whether the target is met, and **path cost** is the energy or time used. Breadth-first search, uniform-cost search or depth-limited search can be used to find the plan.

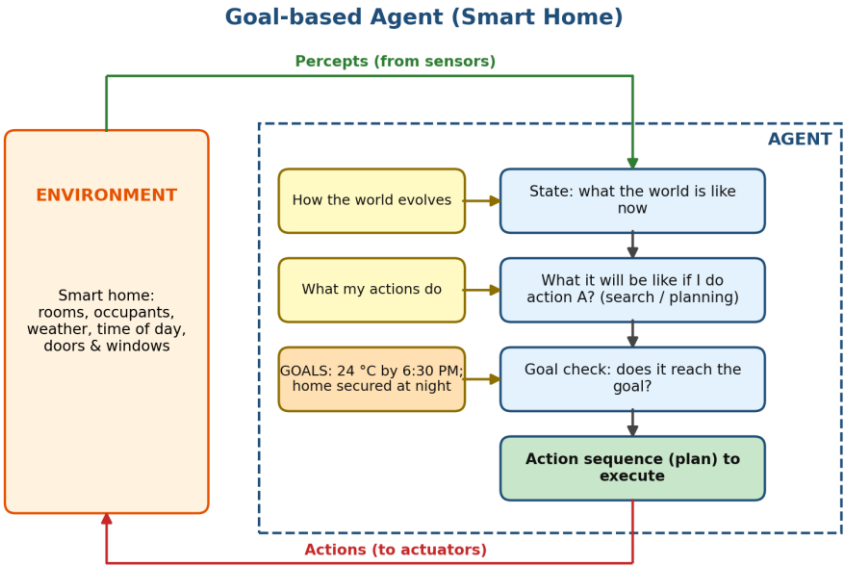


Figure 4: Goal-based agent for the smart home

Working in the smart home:

- **Goal – "Living room at 24 °C when the user arrives at 6:30 PM".** The room is at 32 °C and the AC cools about 1 °C every 5 minutes. The agent plans that 8 °C needs 40 minutes, so it starts the AC at **5:50 PM**.
- **Goal – "Home secured for the night".** Plan: check all doors → lock any unlocked door → close the garage → arm the alarm → switch off non-essential lights.
- **Goal – "Vacation mode".** Plan: lower the thermostat, randomise lights to simulate presence, enable camera alerts.

Advantages: flexible, forward-looking, can pre-condition the house and execute multi-step routines.

Limitations: a goal is binary (achieved or not), so the agent cannot tell which of several plans is better (cheaper, quieter, more comfortable). It cannot trade off conflicting goals such as comfort versus saving, and planning over large state spaces is computationally costly.

2.4 Utility-based Agent

Concept. A utility-based agent uses a **utility function** U that maps each state to a real number measuring how desirable it is. Under uncertainty it chooses the action with the **maximum expected utility**:

$$a^* = \text{argmax}_a \Sigma P(s' \mid a) \cdot U(s')$$

It can compare "good" and "better" outcomes, balance **conflicting objectives**, and make rational choices when outcomes are uncertain.

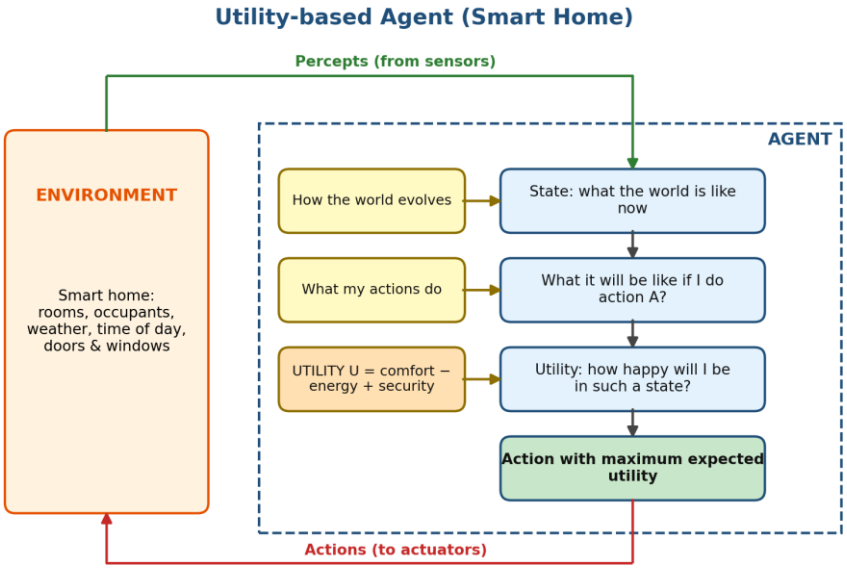


Figure 5: Utility-based agent for the smart home

Working in the smart home. The agent combines the three requirements into one measure:

$$U(s) = w^{c} \cdot \text{Comfort}(s) - w_{e} \cdot \text{EnergyCost}(s) + w_{s} \cdot \text{Security}(s)$$

Worked example – choosing the AC set-point ($w^{c} = 0.6$, $w_{e} = 0.4$, security unchanged):

Action (set-point)	Comfort C	Energy cost E	Utility U = 0.6C – 0.4E
22 °C	0.95	0.90	0.57 – 0.36 = **0.21**
24 °C	0.85	0.55	0.51 – 0.22 = **0.29** (best)
26 °C	0.60	0.25	0.36 – 0.10 = **0.26**

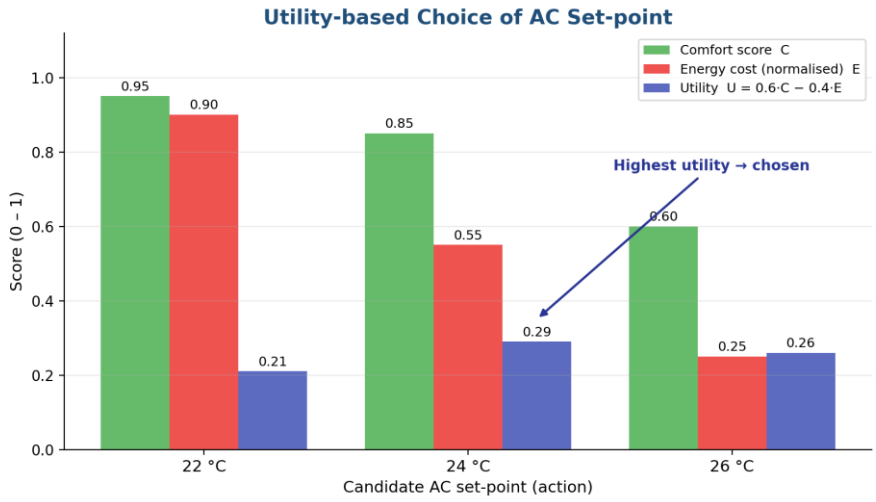


Figure 6: Comparison of utilities for candidate AC set-points

The agent selects **24 °C**: 22 °C is the most comfortable but too expensive, and 26 °C is cheap but uncomfortable. During a **peak-tariff** period the energy weight w_{e} rises, so the optimum shifts toward 25–26 °C automatically. When the house is **empty**, w_{s} and w_{e} dominate, so the agent saves energy

and raises security sensitivity; when someone is **sleeping**, the comfort weight of the bedroom becomes highest.

Advantages: handles trade-offs and uncertainty, produces smooth graded behaviour, and its weights can be learned from user feedback.

Limitations: designing the utility function and weights is difficult, it needs reasonably accurate probability models, and computing expected utility is more expensive than rule matching. It is also too slow to be trusted alone for emergencies such as fire.

3. Comparative Analysis of the Four Agent Types

Criterion	Simple reflex	Model-based	Goal-based	Utility-based
Decision basis	Current percept + IF–THEN rules	Internal state + rules	Goals + look-ahead search	Expected utility of outcomes
Memory / state	None	Yes (world model)	Yes (model + goals)	Yes (model + utility)
Looks ahead?	No	No	Yes	Yes
Partial observability	Poor	Good	Good	Good
Conflicting objectives	Cannot handle	Cannot handle	Limited	Handles well
Learning ability	None	Limited	Moderate	High
Speed / cost	Fastest, cheapest	Fast	Slower (planning)	Slowest (optimisation)
Best smart-home role	Emergency & safety reflexes	Occupancy and device tracking	Scheduling and routines	Comfort–energy trade-off

4. Choosing the Most Effective Agent: A Hybrid Approach

4.1 Why a Single Agent Type Is Not Enough

Requirement of the smart home	Best-suited agent	Reason
Instant reaction to motion, smoke or intrusion	Simple reflex	Millisecond, deterministic, reliable response
Know who is where when sensors are silent or noisy	Model-based	Keeps internal state of the world
Pre-cooling, night mode, away mode, schedules	Goal-based	Plans action sequences to reach a target
Balance comfort, energy cost and security	Utility-based	Quantifies trade-offs
Learn user preferences over time	Learning agent	Improves weights and models from feedback

Each agent type solves only part of the problem. The best solution is therefore a **hybrid, layered, utility-driven agent with a learning component**, shown in Figure 7.

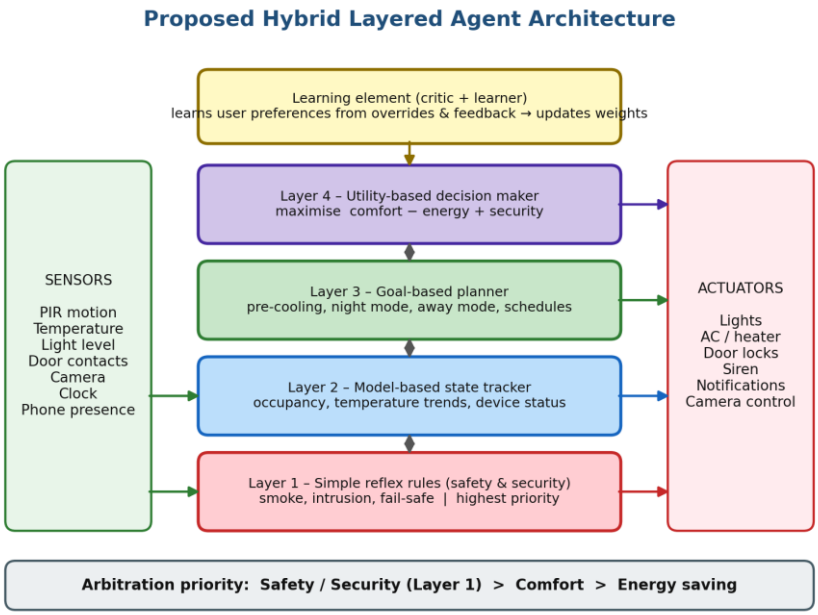


Figure 7: Proposed hybrid layered architecture for the smart home agent

4.2 Layers of the Proposed Architecture

1. **Layer 1 – Reflex layer (safety and security).** Simple IF–THEN rules for smoke, forced entry and sensor failure. It has the **highest priority** and can override every other layer. It also keeps working if the network or the cloud fails (graceful degradation).
2. **Layer 2 – Model-based layer (world model).** Maintains room occupancy, device status, temperature trends and the armed or disarmed state. This shared state is used by all higher layers.
3. **Layer 3 – Goal-based layer (planner).** Converts goals such as “warm house at arrival”, “secure house at night” or “vacation mode” into action sequences using search and planning.
4. **Layer 4 – Utility-based layer (decision maker).** Chooses among alternative plans and settings by maximising U = comfort − energy + security, with weights that depend on context (time of day, tariff, occupancy).
5. **Learning element.** A critic compares results with user feedback (e.g. the user manually sets the AC to 23 °C every evening), and the learning element updates the preference model and utility weights. Over time the system adapts to the household.

Conflict resolution: actions from different layers are arbitrated with the priority **Safety / Security > Comfort > Energy saving**, similar to a subsumption architecture.

4.3 How the Hybrid Agent Meets Each Requirement

Requirement	How the hybrid agent achieves it
Comfort	The model layer tracks occupancy and temperature trends; the goal layer pre-conditions rooms before arrival; the utility layer picks settings that match learned preferences; reflex rules give instant lighting when someone enters.

Requirement	How the hybrid agent achieves it
Energy efficiency	The utility function penalises energy cost and shifts set-points during peak tariff; occupancy tracking prevents lighting or cooling empty rooms; the model layer detects open windows and stops the AC.
Security	Reflex rules respond immediately to intrusion or smoke; the model layer separates resident from unknown presence; the goal layer runs the "secure home" routine; the utility layer raises the security weight when the house is empty or at night.

4.4 Scenario Walk-through

Time	Event	Layer responding and action
5:50 PM	Room at 32 °C, user usually arrives at 6:30 PM (learned)	Goal layer plans 40 min of cooling; utility layer picks an energy-efficient set-point; AC starts.
6:30 PM	User enters; hallway is dark	Reflex layer switches the lights ON instantly.
6:45 PM	User sits still watching TV; PIR silent for 12 min	Model layer infers the room is still occupied, so lights stay on (dimmed).
8:00 PM	Peak tariff begins	Utility layer raises the energy weight and moves the set-point from 24 °C to 25 °C within tolerance.
11:00 PM	Bedtime	Goal layer runs "secure and sleep mode": lock doors, arm alarm, switch off lights.
2:10 AM	Door opens while alarm armed; no resident phone nearby	Reflex layer sounds the siren and sends an alert; the model layer classifies the person as unknown; the utility layer turns on lights and camera.

5. Conclusion

Simple reflex agents give the fast reactions the smart home needs for safety, model-based agents add memory and context in a partially observable world, goal-based agents plan multi-step routines, and utility-based agents resolve conflicts between comfort, energy and security. Each alone is insufficient: reflex agents are blind to context, goal-based agents cannot rank alternatives, and utility-based agents are too slow for emergencies.

A **hybrid layered architecture with a utility-based decision layer, a reflex safety layer and a learning component** is therefore the most effective choice. It reacts instantly to danger, understands the state of the house, plans ahead, optimises the trade-off between comfort and energy, and keeps improving as it learns the habits of the household.

Summary Table

Agent type	One-line role in the smart home
Simple reflex	Instant rule-based safety and security reactions.
Model-based	Tracks occupancy, devices and trends from percept history.
Goal-based	Plans action sequences to reach targets such as pre-cooling and night mode.
Utility-based	Maximises overall satisfaction by trading off comfort, energy and security.

Agent type	One-line role in the smart home
Hybrid + learning	Combines all four with learning, giving the best comfort, efficiency and security.